

N4DIG08 Modbus RTU Protocol(8 Output)

MODBUS command (function code 06 is Control command,03 is Read status command)

Note :

1 MODBUS command must be HEX

2 Slave ID (device address) must be correct, the default slave address is 01, and the Slave ID is set to see the bottom.

3 If you don't remember the Slave ID, use the command Read Slave ID : FF 03 00 FE 00 01 F0 24



As an 8-channel output function, the jumper must be shorted as follows: **M2 1 0**

9600 Band ,8 Data bits,None Parity,1 Stop Bit。

Function code

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16 (2)
	03 Read			
	06 Write			
	16(0x10) Write multiple registers			

Function code	Register address	Register contents	Number of bytes	Register value	Remarks
03 06 16(0X10)	0x0001-0x0008	Output port status	2	0X0000 0X0001	0X0000 Relay Close 0X0001 Relay Open
03 06	0x00FA (250)	Onput level	2	0X0000 0X0001	0X0000 PNP Low level output (default) 0X0001 NPN High level onput
03 06	0x00FE	Baud rate	2	0x0000- 0x0005	0~5 0:1200 1:2400 2:4800 3:9600 (default) 4:19200 5: Factory reset
03 06	0x00FF	Baud rate	2		0~4 0:1200 1:2400 2:4800 3:9600 (default) 4:19200 5: Factory reset

MODBUS 06 Command (Control command ,HEX):

Bytes Number	1	2	3	4	5	6	7	8
MODBUS Definitions	Slave ID	Function	Address		Data		CRC Check	
Function	Device Address	Function	Channel number		Command	Delay time	CRC Check	
Open	0x00-0x2F	0x06	0x0001-0x0008		0x01	0x00	2Bytes CRC	
Close	0x00-0x2F	0x06	0x0001-0x0008		0x02	0x00	2Bytes CRC	
Toggle (Self-locking)	0x00-0x2F	0x06	0x0001-0x0008		0x03	0x00	2Bytes CRC	
Latch Inter-locking)	0x00-0x2F	0x06	0x0001-0x0008		0x04	0x00	2Bytes CRC	
Momentary (Non-locking)	0x00-0x2F	0x06	0x0001-0x0008		0x05	0x00	2Bytes CRC	
Delay	0x00-0x2F	0x06	0x0001-0x0008		0x06	0x00-0xff	2Bytes CRC	
Open all	0x00-0x2F	0x06	0x0000		0x07	0x00	2Bytes CRC	
Close all	0x00-0x2F	0x06	0x0000		0x08	0x00	2Bytes CRC	

Remarks:

1 Momentary mode, delay time is 1 seconds

2 Delay mode, delay time is 0-255 seconds

Return command:

Command is active, return to send commands; instruction is invalid no return.

MODBUS 03 Command (Read status command ,HEX):

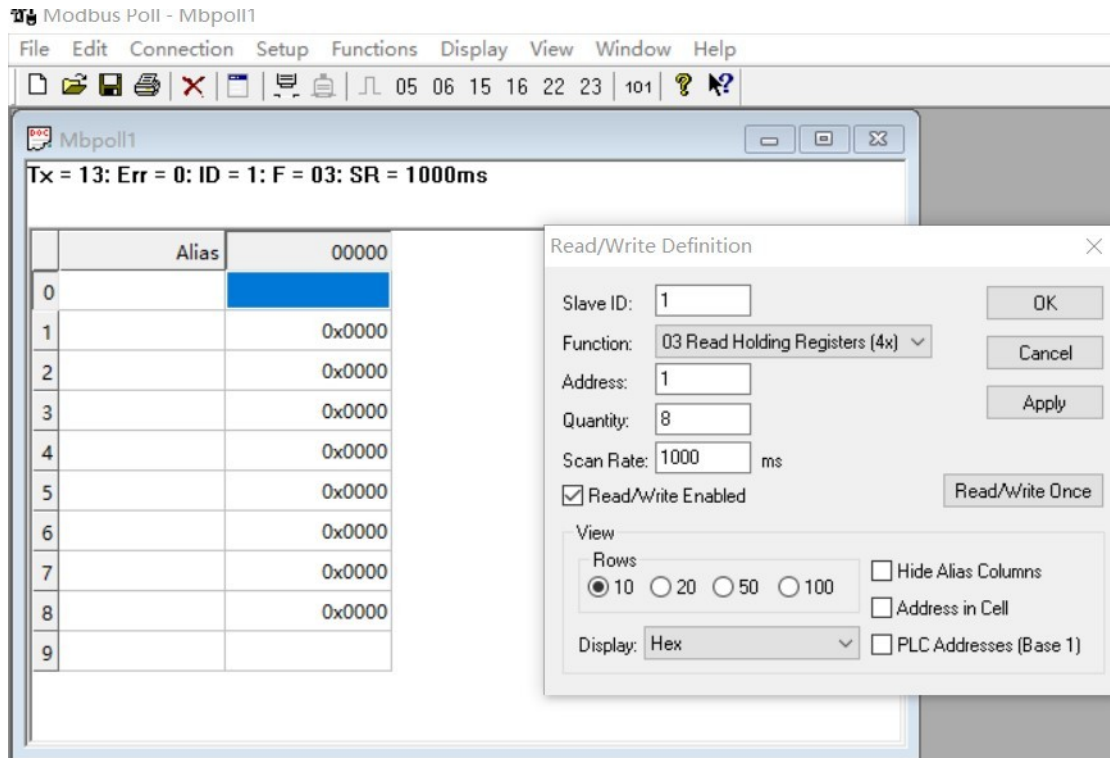
Bytes Number	1	2	3	4	5	6	7	8
MODBUS Definitions	Slave ID	Function	Address		Data		CRC Check	
Function	Device Address	Function	Starting register address		Register length		CRC Check	
Read Channel 1 State	0x00-0x2F	0x03	0x0001		0x0001			
Read Channel 2 State	0x00-0x2F	0x03	0x0002		0x0001			

Read 2 consecutive channels status	0x00-0x2F	0x03	0x0001-0x0003	0x0002	
Read 3 consecutive channels status	0x00-0x2F	0x03	0x0001-0x0002	0x0003	
Read all 8 channels status	0x00-0x2F	0x03	0x0001	0x0008	

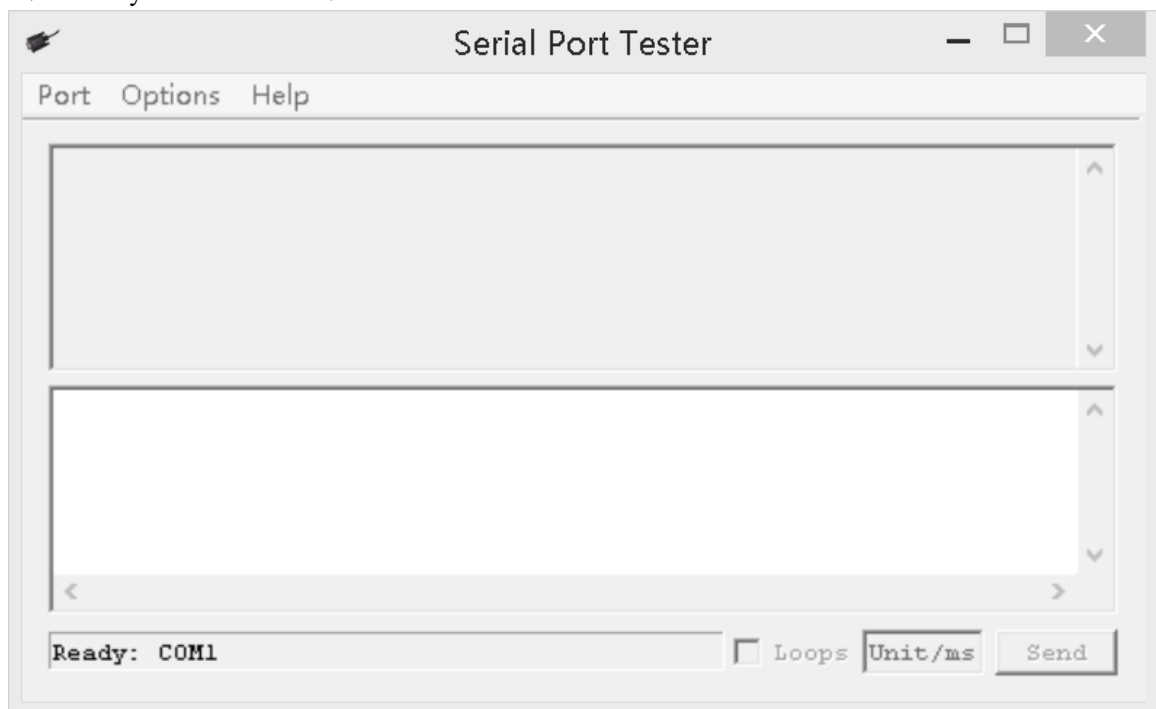
Read status command returns (function code 03, HEX format):

Bytes length	1	1	1		2
MODBUS Definitions	Slave ID	Function	data length	data	CRC16 Check
Function	Device Address	Function	data length	Relay state 0x0001 open 0x0000 close	CRC16 Check
Channel 1 open	0x00-0x1F	0x03	0x02	0x0001	
Channel 1 close	0x00-0x1F	0x03	0x02	0x0000	
Channel 2 open	0x00-0x1F	0x03	0x02	0x0001	
Channel 2 close	0x00-0x1F	0x03	0x02	0x0000	
Channel 1 open Channel 2 open	0x00-0x1F	0x03	0x04	0x0001 0x0001	
Channel 1 open Channel 2 close	0x00-0x1F	0x03	0x04	0x0001 0x0000	
Channel 1 close Channel 2 open	0x00-0x1F	0x03	0x04	0x0000 0x0001	
Channel 1 close Channel 2 close	0x00-0x1F	0x03	0x04	0x0000 0x0000	

MODBUS commands you can use "Modbus Poll" input, as shown below
(CRC check generated automatically)



You can also use HyperTerminal serial input, as shown below
(Manually add CRC check)



Examples

Channel 1 Open : 01 06 00 01 01 00 D9 9A

Channel 1 Close : 01 06 00 01 02 00 D9 6A

Channel 1 Toggle: 01 06 00 01 03 00 D8 FA
Channel 1 Latch: 01 06 00 01 04 00 DA CA
Channel 1 Momentary: 01 06 00 01 05 00 DB 5A
Channel 1 Delay 10 seconds : 01 06 00 01 06 0A 5B AD
Channel 1 Delay 100 seconds: 01 06 00 01 06 64 DA 41

Channel 2 Open : 01 06 00 02 01 00 29 9A
Channel 2 Close : 01 06 00 02 02 00 29 6A
Channel 2 Toggle : 01 06 00 02 03 00 28 FA
Channel 2 Latch : 01 06 00 02 04 00 2A CA
Channel 2 Momentary : 01 06 00 02 05 00 2B 5A
Channel 2 Delay 10 seconds : 01 06 00 02 06 0A AB AD
Channel 2 Delay 100 seconds : 01 06 00 02 06 64 2A 41

Read state (assuming that the channel 1 is open, the channel 2 is close).

Read channel 1 state : 01 03 00 01 00 01 D5 CA

Return open: 01 03 02 00 01 79 84

Read channel 2 state : 01 03 00 02 00 01 25 CA

Return close: 01 03 02 00 00 B8 44

Read channel 1 and channel 2 state : 01 03 00 01 00 02 95 CB

Return channel open and channel 2 close : 01 03 04 00 01 00 00 AB F3

1. Read Output level selection register

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16 (2)
--	--------------	-------------------------	-----------------------	---------------

Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16 (2)
--	--------------	------------------------	----------	---------------

RS485 address: 0x01-0xFE

Function code 0x03

Register address: 0x00FA

Read number: 0x0001

For example:

send data(RS485 address is 1): 01 03 00 FA 00 01 A4 3B

Returns data 1 : 01 03 02 00 00 B8 44 PNP low level output (default)

Returns data 2 : 01 03 02 00 01 79 84 NPN high level output

2. Set Output level selection register (8 channels set at the same time)

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Setting Content (2)	CRC16(2)
--	-----------------	-------------------------	------------------------	--------------

Returns data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Register value (2)	CRC16(2)
--	-----------------	----------------------------	-----------------------	--------------

RS485 address:0x01-0xFE

Function code 0x06

Register address: 0x00FA

Setting Content: 1Bytes

0X0000 Low level output (default): Low level is valid output, high level is invalid output

0X0001 High level output: high level valid output, low level invalid output

Set low level input(PNP):

Send data (RS485 address is 1) 01 06 00 FA 00 00 A9 FB

Return data: 01 06 00 FA 00 00 A9 FB

Set high level input(NPN):

Send data (RS485 address is 1) 01 06 00 FA 00 01 68 3B

Return frame: 01 06 00 FA 00 01 68 3B

3. Read RS485 address

Send data

RS485 address (Broadcast address) (1)	Function (1)	Register address (2)	Read (2) number	CRC16(2)
--	-----------------	-------------------------	--------------------	--------------

Returns data

RS485 address (Broadcast address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
--	-----------------	------------------------	----------	--------------

Broadcast address 0xff

Function code 0x03

Register address: 0x00FE

Read number: 0x0001

For example:

send data: FF 03 00 FE 00 01 F0 24

Returns data: FF 03 02 00 01 50 50

FF Broadcast address, 03 Function, 02 length, 01 is the current module

RS485 address , 50 50 crc16

Note: When using this command, only one temperature module can be connected to the RS485 bus, more than one will be wrong!

4. Write RS485 address

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Setting Content (2)	CRC16(2)
--	-----------------	-------------------------	------------------------	--------------

Returns data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Register value (2)	CRC16(2)
--	-----------------	----------------------------	-----------------------	--------------

RS485 address (Slave ID) : 0x01~0xFE

Function code 0x06

Register address: 0x00FE

Setting Content: 2Bytes(1-247)

For example, The current RS485 address is 1, We need to change the RS485 address to 3:

send data(RS485 address is 1): 01 06 00 FE 00 03 A8 3B

Returns data: 01 06 00 FE 00 03 A8 3B

5. Read baud rate

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Read number (2)	CRC16(2)
--	-----------------	-------------------------	-----------------------	--------------

Returns data

RS485 address (Station address) (1)	Function (1)	Number of bytes (1)	data (n)	CRC16(2)
--	-----------------	------------------------	----------	--------------

RS485 address (Slave ID) : 0x01~0xFE

Function code 0x03

Register address: 0x000FF

Read number: 0x0001

For example:

send data(RS485 address is 1): 01 03 00 FF 00 01 B4 3A

Returns data: 01 03 02 00 03 F8 45

01 RS485 address, 03 Function, 02 length, F8 45 crc16

03 means the current baud rate is 9600bps

Baud rate corresponds to the number: 0: 1200 1: 2400 2: 4800 3: 9600 4: 19200

6. Write RS485 address

Send data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Setting Content (2)	CRC16(2)
--	-----------------	-------------------------	------------------------	--------------

Returns data

RS485 address (Station address) (1)	Function (1)	Register address (2)	Register value (2)	CRC16(2)
--	-----------------	----------------------------	-----------------------	--------------

RS485 address (Slave ID) : 0x01~0xFE

Function code 0x06

Register address: 0x00FF

Setting Content: 2Bytes(0-4)

For example, Change the baud rate to 4800bps:

send data(RS485 address is 1): 01 06 00 FF 00 02 38 3B

Returns data: 01 06 00 FF 00 02 38 3B

Baud rate corresponds to the number: 0: 1200 1: 2400 2: 4800 3: 9600 4: 19200

5: Factory reset

Note: 1 The baud rate will be updated when the module is powered up again!

2 The factory setting can be restored when the baud rate corresponding to the number is 5. For example: 01 06 00 FF 00 05 79 F9